STC-Net: Scattering Topology Cue-Based Network for Aircraft Detection in SAR Images

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***Abstract*— Aircraft detection in synthetic aperture radar (SAR) imagery is significant due to its critical role in various applications, including surveillance, reconnaissance, and security. However, given the background interference and discreteness of aircraft scattering, detectors are prone to acquire unremarkable aircraft features. These factors lead to false alarms and present difficulties in locating aircraft accurately. This article proposes an innovative scattering topology cue-based network (STC-Net), which enhances aircraft discriminability and more accurately evaluates the quality of the prediction results. We model the aircraft with the star topology (ST), which not only emphasizes critical components like the nose and wings but also explicitly links them as a cohesive unit. Based on the cue of ST, the ST space fusion module (ST-SFM) and the ST channel attention module (ST-CAM) are designed. The former integrates discrete components to reestablish the aircraft features based on neighboring information of ST, while the latter suppresses background interference to highlight the aircraft by exploiting node information of ST. In addition, completeness and consistency loss (CCLoss) function that includes the completeness-aware label and the positive sample weighting function is introduced. The completeness-aware label describe the localization accuracy by incorporating the degree of overlap of predicted results on ST, while the positive sample weighting function enhances the consistency of the classification and localization branches. Furthermore, experiments conducted on the Gaofen-3 SAR aircraft detection dataset (GF3ADD) and the publicly available SAR-AIRcraft-1.0 dataset demonstrate the effectiveness and generalizability of STC-Net, with our method achieving state-of-the-art performance.**

***Index Terms*— Completeness, consistency, scattering topology, star topology (ST), synthetic aperture radar (SAR) aircraft detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor that operates continuously, regardless of weather conditions, and has been utilized in numerous applications [1], [2], [3], [4]. As a fundamental task in SAR image interpretation, aircraft detection serves many downstream applications [5], [6], [7], such as intelligence reconnaissance and air traffic control.

SAR image object detection algorithms have been evolving rapidly over the past decades. Traditional methods are primarily classified into two categories: grayscale-based models and geometrical-based models. By modeling the background, constant false alarm rate (CFAR) [8], [9], [10], a typical representative of grayscale-based models, obtains the threshold used to distinguish targets from the background under the condition of CFAR. To achieve the modeling of inhomogeneous backgrounds, many statistical models have been proposed [9], [11], [12], such as G^0 distribution. In addition, some works are devoted to introduce spatial distribution as auxiliary information in CFAR detectors to reduce false alarms, such as the bilateral CFAR detector [13]. As the resolution becomes finer, more geometric features of the target are presented in the SAR images, which supports the advancement of geometrical-based models. The aircraft skeleton and other components are individually extracted by Hough transform [14], covariance, and symmetry to detect the aircraft. Canny operator [15] and active contour model [16] are also commonly used in detectors to obtain geometric structure information such as target edges. In conclusion, traditional methods can be competent for detection in simple and specific scenarios with the help of the priori information hand-crafted features, but generalization ability is insufficient due to their heavy reliance on such factors.

Benefiting from the powerful ability to extract feature representations, deep learning-based models have achieved outstanding results in many domains [17], [18], [19], [20]. And abundant SAR images are more readily available [21], creating an auspicious environment to develop deep learning on SAR images. As one of the initial attempts, autoencoder extracts features from raw images to train a soft-max classifier [22]. The saliency-based predetection method is combined with CNN to generate region suggestions and refine localization and classification [23]. This idea of combining deep learning with traditional machine learning is also reflected in [24], [25], [26], and [27]. With this simple combination, deep learning alleviates the reliance of hand-crafted features. To enhance efficiency, end-to-end detectors have been proposed for SAR image aircraft detection. Some general and powerful modules are also present in the models. For example, the feature pyramid module improves the performance of

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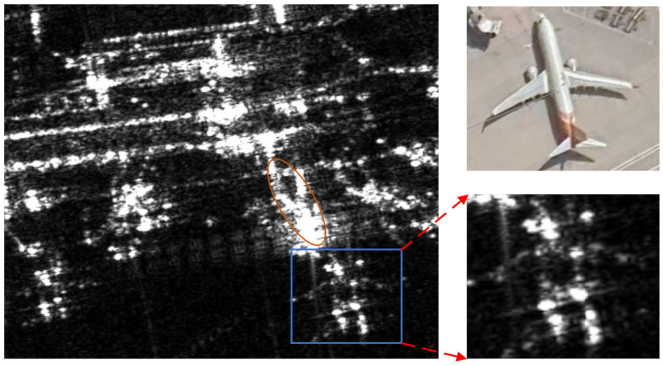


Fig. 1. Aircraft and artificial facilities in airport. The main building is the terminal, with the blue box and the orange circle representing the aircraft and the corridor, respectively.

detecting aircraft at different scales [5]. A single-shot detector [28] refines and aligns features for aircraft detection by the attention mechanism. These methods represent an extension of deep learning paradigms and have demonstrated superior performance.

However, in view of the characteristics of SAR images, certain challenges remain and warrant focused attention for effective SAR aircraft detection. The wavelength of the SAR signal exceeds that of visible light, resulting in observed targets that exhibit discrete and multicentric characteristics. As structurally intricate entities, aircraft manifest greater discreteness as shown in Fig. 1, and their fragmented features constrain the efficacy of detection algorithms primarily designed for processing continuous, intricate features. In particular, artificial facilities such as corridors and bridges in complex airport environments demonstrate similar characteristics, posing challenges for the model to distinguish aircraft from the background. Consequently, suppressing false alarms becomes a difficult task. In addition, since SAR is the coherent imaging system, inherent factors such as speckle noise and sidelobes inevitably emerge, which tend to blur target edge contours and weaken target texture information. This phenomenon exacerbates the difficulty in accurately identifying aircraft. Overall, aircraft features in SAR images are discrete and susceptible to interference from factors such as complex backgrounds and noise. However, most of the existing detection methods uniformly focus on the aircraft region and exhibit limited capability in associating discrete and fragmented target features. This limitation exacerbates the difficulty in accurately mapping multiple scattering centers to a single aircraft, consequently constraining the upper limit of performance.

To solve the above problems, an innovative scattering topology cue-based network (STC-Net) is proposed for aircraft detection in SAR images. Aiming to refine aircraft modeling, STC-Net constructs a star topology (ST) by connecting discrete scattering centers according to spatial relationships. The ST is able to associate multiple scattering centers into a single aircraft in complex backgrounds, which helps the model to extract aircraft semantic information accurately. Specifically, the ST space fusion module (ST-SFM) utilizes the neighboring information of the ST to dynamically aggregate the discrete scattering centers, obtaining more complete aircraft features. The ST channel attention module (ST-CAM) leverages node information of the ST to enhance aircraft saliency in the presence of strong interference. ST-SFM and ST-CAM based on ST grant higher attention to critical scattering regions and overcome discreteness in the case of disturbances such as noise for efficient extraction of aircraft features. Additionally, a completeness and consistency loss (CCLoss) is formulated for aircraft with discrete, multiple scattering centers. Without limiting itself to continuous and complete targets, CCLoss could guide the model to obtain predictions that more perfectly cover the ST, which means more accurate aircraft localization. This makes the STC-Net to perform better with complex backgrounds and discrete features. In summary, from aircraft modeling and feature enhancement to the loss function, the ST provides comprehensive guidance for key steps in the detection. This makes STC-Net more suitable for aircraft detection under complex interference, compared to existing methods designed for processing continuous, intricate objects. The main contributions of our work are summarized as follows.

1) Combined with the scattering characteristics of SAR aircraft, the ST is proposed to consolidate multiple scattering centers into a single aircraft. ST enables critical scattering regions, such as the nose and wings, to receive more attention while reinforcing their association. As a result, the STC-Net, which is structured upon structural cues from ST, exhibits superior detection performance in handling extreme features and complex backgrounds.
2) To enhance aircraft saliency, the ST-SFM and the ST-CAM are proposed. The ST-SFM dynamically adjusts the receptive field by leveraging neighboring information of ST to aggregate discretely distributed aircraft features. Meanwhile, ST-CAM generates a structural importance-based channel mask using the node information of ST to suppress background interference.
3) To improve the model's capacity to measure the completeness of prediction results and bolster the alignment between classification and regression tasks, CCLoss is introduced. CCLoss gauges the completeness of samples by leveraging the spatial distribution of the ST. Moreover, during training, it prioritizes attention toward inconsistent samples, thus fostering improved model performance.

Experiments are performed on Gaofen-3 SAR aircraft detection dataset (GF3ADD), demonstrating the superior performance of STC-Net over alternative methods. And the following content is organized as follows. In Section II, we provide a brief review of related work. Section III presents the proposed methodology in detail. Section IV offers detailed descriptions of the experiment setup and results. Finally, conclusions are drawn in Section V.

II. Related Work

A. General Detection Models

Deep convolutional neural networks [29], [30] have achieved significant success since their inception, thanks to

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their robust feature extraction capabilities. Currently, deep learning algorithms for detection are mainly categorized into anchor-based and anchor-free methods.

Anchor-based detection algorithms can be subdivided into two primary categories: two-stage detectors and one-stage detectors. And the R-CNN series [31], [32], [33] serves as prominent examples of two-stage detectors. In the same period of work, SPPnet [34] uses spatial pyramid pooling to implement shared computation, thus achieving an increase in detection speed. Mask R-CNN [35] proposes the ROI alignment module and the mask prediction branch while incorporating the instance segmentation task. Cascaded R-CNN [36] trains a multistage detector with higher detection accuracy by resampling samples and gradually increasing the quality of positive samples. To address the compromise between detection accuracy and processing speed, the researchers propose a series of one-stage detectors. The YOLO series [37], [38] exemplifies single-stage detectors, offering real-time efficiency for end-to-end prediction. SSD [39] and FPN [40] leverage multiscale feature maps to address the challenges arising from variations in objects scales. RetinaNet [41] proposes focal loss (FL), which assigns different loss weights to samples according to their level of training difficulty.

To avoid introducing anchor-related hyperparameters and calculations, anchor-free methods have been proposed, which include two main categories. One kind of algorithm is based on the keypoints. CornerNet [42] models objects as paired keypoints and localizes them by predicting the corner points. ExtremeNet [43] detects four extreme points to determine target boundaries and aggregates them by detecting centroids. CenterNet [44] and CSP [45] models the objects as centroids and then determines the location by regressing the length and width of the objects. The other type is dense prediction algorithms. DenseBox [46] predicts the probability and regresses the distance to the ground truth for each position on the feature map, without using the hand-designed anchor. According to the scale, FCOS [47] detects the object on the corresponding feature map, and proposes center-ness branch to filter the prediction results away from the object center. FoveaBox [48] increases the distinction between samples by dividing the sampling area.

B. Aircraft Detection Models in SAR Images

Traditional methods detect aircraft by modeling background clutter or relying on hand-crafted features, which are often insufficient for complex scenarios. Thanks to the powerful representation and generalization capabilities, deep learning-based methods [49], [50] are widely utilized for SAR aircraft detection. Zhao et al. [51] construct a global instance dataset to enhance interclass variability and intraclass consistency to improve the performance of classification branching. Zhou et al. [52] introduce the diffusion model to achieve precise positioning by denoising the bounding box. Some approaches introduce attention mechanisms that combine SAR images and aircraft characteristics to aircraft detectors. Considering that aircraft are sensitive to imaging angles, a spatial orientation attention network [53] is proposed to capture the feature change patterns of aircraft under different orientations. Chen et al. [54] use the geographic information attention module in their model for aircraft target detection. Similarly geographic information is used in [55] and [56] to improve aircraft detection efficiency and reduce false alarms as auxiliary information. Zhao et al. [5] propose a dilated attention block to enhance the internal association of aircraft targets. To detect multiscale aircraft targets, the feature pyramid structure is introduced and optimized in the detectors. An et al. [57] model aircraft with rotated boxes and use the feature pyramid network to improve detectability of small-scale aircraft. Ye et al. [58] combines features from critical parts of an aircraft to enhance multiscale features. Xiao et al. [59] perform peak feature enhancement on the image and later fuse the multiscale features using adaptive deformable convolution (DC). Zhang et al. [60] use scale expansion and feature enhancement to construct feature pyramids. Further, researchers use scattering information to guide the construction and training of the model. Guo et al. [6] align and model the strong scattering points using a hybrid Gaussian model and then enhance the images. Kang et al. [61] enhance the strong scattering point features and strengthen the relationship between strong scattering points to generate complete detection results for aircraft. By leveraging scattering points, Chen et al. [62] eliminate the reliance on anchors while enhancing the semantic information pertaining to the aircraft. Wang et al. [63] propose a constrained semantic conditional constraint-guided feature-aware network using information such as aircraft morphology.

While these methods leverage the properties of SAR images to enhance detection performance to a certain extent, they inadequately address the effect of discrete aircraft features on model performance, particularly under complex interference. Therefore, we propose ST to associate discrete multiple scattering centers into a unified aircraft target. STC-Net leverages cues from the ST to inform both aircraft feature extraction and model optimization processes, improving detection performance.

III. Proposed Method

To mitigate false alarms and improve aircraft localization accuracy, we propose STC-Net, representing an aircraft by ST, which not only emphasizes critical components like the nose and wings but also explicitly links them as a cohesive unit. Its core is to utilize the structural cues of ST to enhance the saliency of the aircraft features and guide the model to accurately evaluate the prediction results. The overall structure of the STC-Net is shown in Fig. 2, comprising three main components: the backbone with FPN, the star topology feature fusion module (STFFM), and the completeness-aware detection head. In the subsequent part of this section, more details of STC-Net are described.

A. Preliminary Network Structure

Anchor-free methods bypass predefined anchor boxes, thereby circumventing the introduction of hyperparameters like sizes and aspect ratios, which is suitable for detecting SAR

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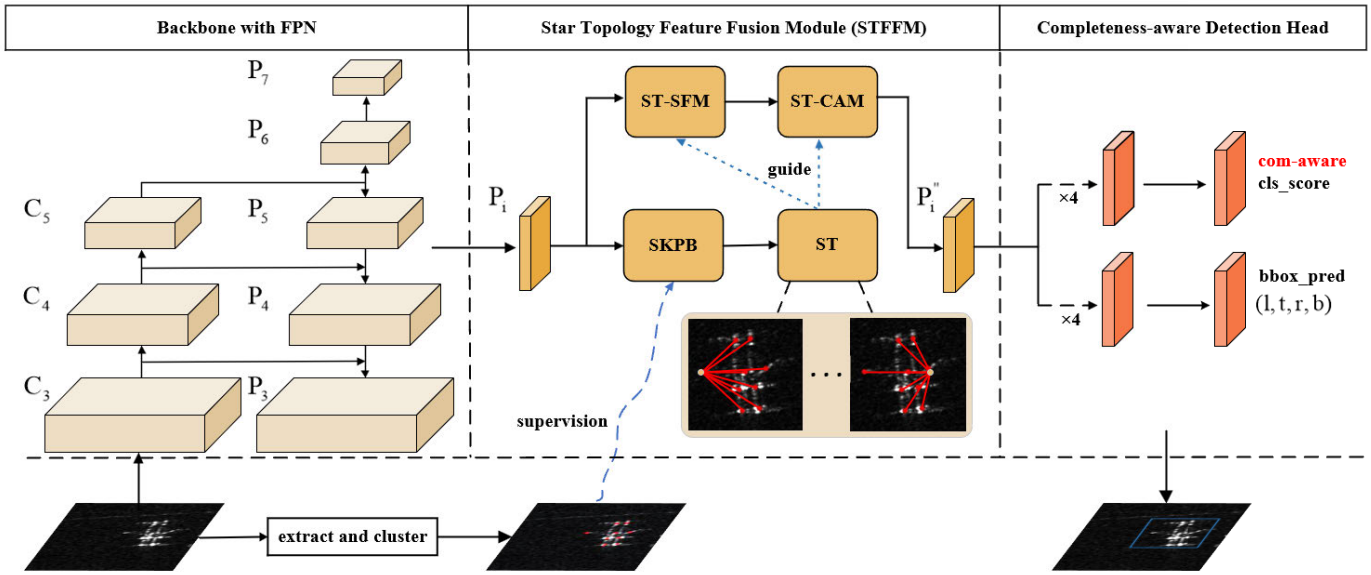


Fig. 2. Overall structure of the STC-Net, comprising three main components: the backbone with FPN, the STFFM, and the completeness-aware detection head. The SAR image is processed through the backbone with FPN to obtain initial features. Subsequently, the initial features are concurrently fed into the SKPB and the STFFM. Based on the predictions of SKPB, the ST is constructed. Specifically, the neighboring information is used to guide the ST-SFM, while the node information is utilized to guide the ST-CAM. Passing through the STFFM, the enhanced features are obtained and then input into the detection head.

aircraft with large-scale variations and sparse distribution. Therefore, the proposed methodology references the network structure of FCOS, an anchor-free detector. In addition, FCOS assigns targets to different resolution feature maps for detection based on their respective scales, which facilitates detecting SAR aircraft at different scales efficiently.

First, the image is passed through ResNet50 to extract features, yielding feature maps $\{C_3, C_4, C_5\}$, where each map has downsampling rates of $\{8, 16, 32\}$, respectively. Subsequently, the feature maps are inputted into FPN. Feature maps P_i with downsampling rates of 2^i are obtained after FPN, where $i \in \{3, 4, 5, 6, 7\}$. However, as generalized networks designed for natural scenes, neither ResNet50 nor FPN consider the discreteness of aircraft scattering and the complexity of background interference in SAR images. Consequently, we exploit the structural information inherent in scattering keypoints to design STFFM, which comprises two sub-modules: ST-SFM and ST-CAM. And the feature map P_i is further enhanced to obtain more salient aircraft features. Finally, the detection head is connected to the feature maps for classification and localization. Unlike FCOS, the detection head of STC-Net omits the centerness branch, which typically addresses the decline in detection performance resulting from predictions deviating from the center of the target. In its place, CCLoss is formulated to steer the training process, bolstering the model's capability to discern localization accuracy, thus filtering out the results of inaccurate localization.

The above describes the structure of the model and the aircraft inspection process, and the specific innovative design is described in detail in the subsequent parts.

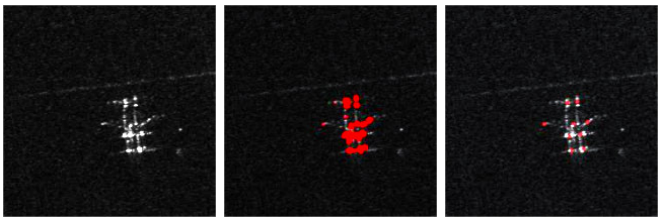
B. Scattering Keypoint Prediction

The nose, wings, and engines of the aircraft have stronger backscattering, manifesting as brighter yet distinct points in SAR images. These conspicuous scattering points can delineate the aircraft's structure. In addition, the aircraft in SAR images lacks detailed information such as color and texture, making them less visually distinguishable. Therefore, it is necessary to introduce scattering keypoints as auxiliary supervised signals in the training, which can help the model to learn representative features in a descriptive manner.

The strong scattering keypoints are obtained by the following steps. Due to the inherent speckle noise in SAR images, which causes random fluctuations in pixel values, the image is first processed through a mean 5×5 filter to mitigate the effects of speckle noise. Second, considering the scattering characteristics of each part of the airplane, we use different ways to extract the scattering points. For wings and tails, which primarily involve edge wrapping, the Canny algorithm is employed to extract edge information. Conversely, for the nose and engines, which predominantly entail dihedral and trihedral angle reflections, the Harris algorithm is utilized to extract corner point information. The scattering points of the aircraft consist mainly of two types of points mentioned above, as shown in Fig. 3(b). The extraction range is limited to the ground-truth bounding box during train, which helps eliminate unnecessary interferences, such as those from terminals. After suppressing these high-intensity interferences, the aircraft target is highlighted, facilitating the extraction of corner and edge information. Finally, centers are obtained by k-means clustering algorithm, as shown in Fig. 3(c), which are usually distributed in the positions of nose and wing. M scattering point centers are used as ground truths of the aircraft's scattering keypoints. It should be noted that the value of M is constrained by ST-SFM, with corresponding explanations provided in its detailed description.

In the training, the scattering keypoints prediction branch (SKPB) is proposed to predict the location of scattering

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(a) (b) (c)

Fig. 3. Extraction process of scattering keypoints. (a) Original image. (b) Image processed by Canny and Harris algorithms. (c) Image processed by k-means clustering algorithm.

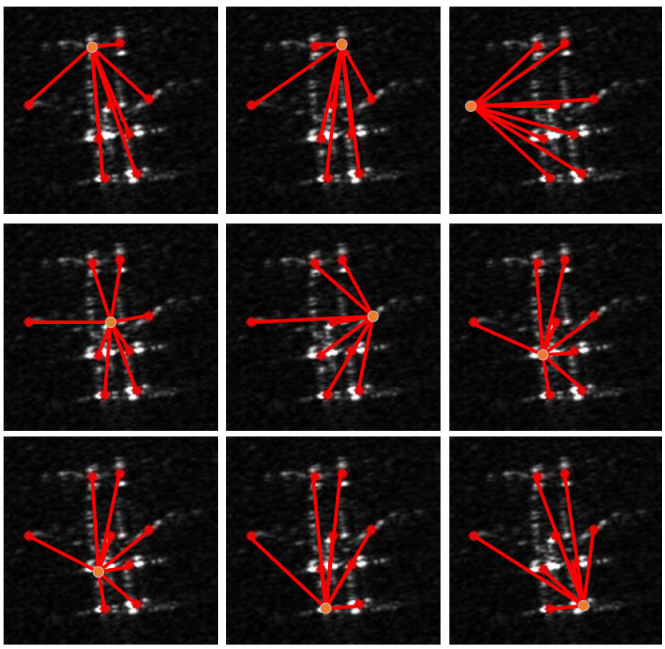


Fig. 4. ST. M star topologies are constructed centered on each scattering keypoint within the aircraft, where M denotes the number of scattering keypoints.

keypoints. Within SKPB, the feature map P_i is first passed through stacked four 3×3 convolution layers, and then downscaled through a 3×3 convolution layer to obtain the heatmap H_i of the scattering keypoints. The channel dimension of H_i is 1 and the resolution is consistent with P_i. FL is used as the objective function to supervise the training of SKPB, as depicted in

$$\mathrm{FL}\left(p^{\mathrm{sk}}, q^{\mathrm{sk}}\right) = \begin{cases} -\left(1-p^{\mathrm{sk}}\right)^{\gamma} \log p^{\mathrm{sk}}, & q^{\mathrm{sk}} = 1 \\ -\alpha_1 \left(p^{\mathrm{sk}}\right)^{\gamma} \log\left(1-p^{\mathrm{sk}}\right), & q^{\mathrm{sk}} = 0 \end{cases} \quad (1)$$

where p^{sk} denotes the prediction probability and q^{sk} denotes whether the location is a scattering critical point or not. And the corner marker sk is an abbreviation for scattering keypoints. As moderators, α_1 is set to 0.25 and γ is set to 2, according to FL.

C. ST Feature Fusion Module

Strong background interference and speckle noise pose a great challenge for the model to extract aircraft features. Meanwhile, the aircraft comprises discrete strong scattering points with weak correlation. And the scattering keypoints contain information about the geometry of the aircraft, which is key to distinguishing the aircraft from the background. Hence, combining structural information contained in the scattering keypoints to obtain more discriminating features is very important. At each scattering keypoint, we build a ST centered on it to re-represent the aircraft, as shown in Fig. 4. Inspired by the star-shaped box proposed in VarifocalNet [64], the proposed ST is based on scattering keypoints, reflecting the structural characteristics of airplanes. Based on this, the STFFM is proposed, which includes ST-SFM and ST-CAM.

1) Star Topology: ST consists of two primary components: node information, which records the attributes of the entities constituting ST, and neighboring information, which contains the spatial relationships among these nodes. They are all derived from the node positions, and the set of node positions is denoted as $S = \{s_1, s_2, \ldots, s_M\}$, where $s_i = (x_i, y_i)$ in the set represents the 2-D coordinate of the node. For SAR aircraft, nodes represent critical components with strong backscattering intensity, such as the nose and wings, and s_i is consequently derived by scattering keypoint prediction.

2) ST Space Fusion Module: Scattering keypoints correspond to the components of the aircraft that exhibit strong backscattering intensity. By using the positional relationship between them, the structure of the aircraft can be completely described. However, due to constraints imposed by the fixed sampling method and limited receptive field, traditional convolution operations struggle to adequately cover critical scattering regions. Therefore, leveraging the neighboring information of ST to direct DC [65], we propose the ST-SFM to obtain the complete aircraft features, as shown in Fig. 5.

The neighboring information of ST is represented as relative coordinates between nodes in the ST-SFM. First, the feature P_i undergoes cascading 3×3 and 1×1 convolution operations to obtain the initial offset, as in [65]. Second, the relative coordinates of the nodes are calculated. For instance, for the node s_i, the relative coordinates of the other scattering keypoints are R, as depicted in the following equation:

$$R = \{(x_1 - x_i, y_1 - y_i), \ldots, (x_{i-1} - x_i, y_{i-1} - y_i), \ldots, (x_M - x_i, y_M - y_i)\}. \quad (2)$$

Relative coordinates are then used to refine the initial offset at the position corresponding to the node s_i. Finally, DC is performed on the feature map P_i based on the refined offsets to obtain P_i'. From the above process, it is evident that the number M of scattering points is determined by the size of the convolution kernel, specifically M equals the square of the size of the convolution kernel. In typical scenarios, the convolution kernel size is 3, hence M is set to 9.

To refine the structural features of the airplane, DC is utilized in [28], which results in improving the performance of the detector. However, the offset obtained is poorly interpretable due to the absence of an explicit supervisory signal. Unlike the way they obtain deformable convolutional offsets directly from vanilla convolutions, we achieve this through the prediction of scattering keypoints and the ST of the airplane. By ST-SFM, the model aligns the features of the region where the scattering keypoints are located. The model not only

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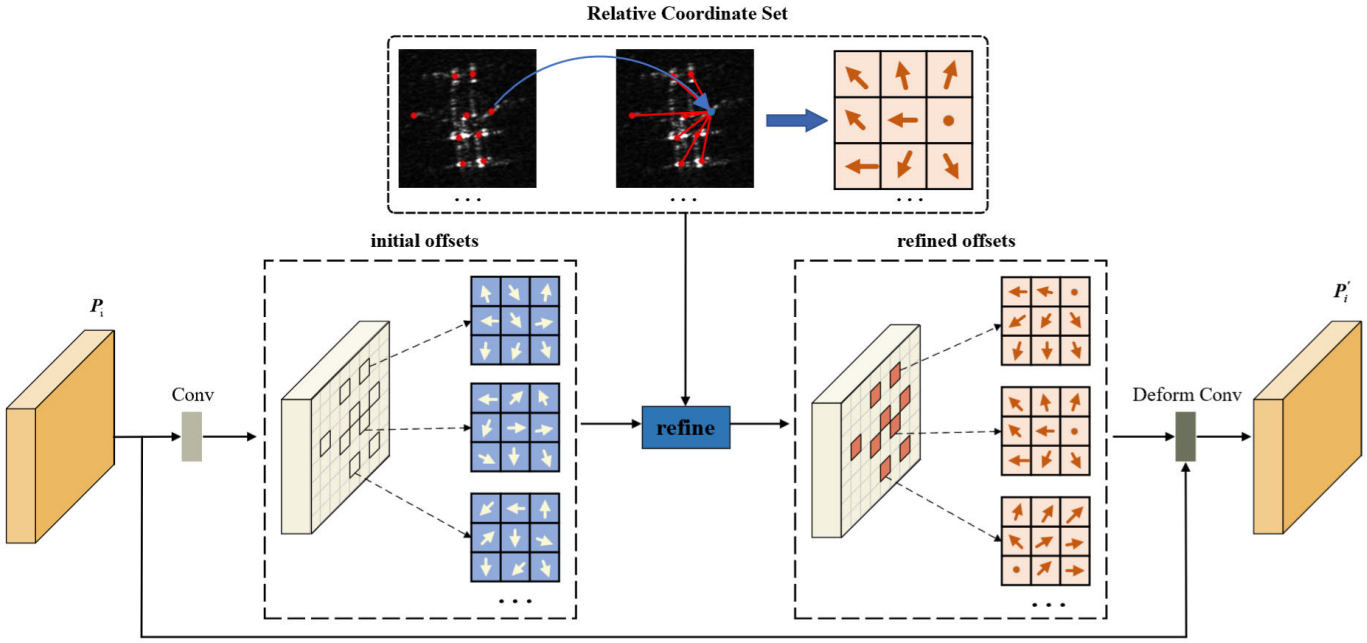


Fig. 5. ST-SFM. The feature map undergoes convolution to obtain initial offsets. Subsequently, the values at the locations of the scattering keypoints are replaced by the relative coordinate set, which represents the neighboring information in the ST. The refined offsets are then used to guide the DC.

learns the geometric structural features of the aircraft, but also mitigates the incompleteness of the aircraft in the SAR image.

3) ST Channel Attention Module: The complex background of the airport and the speckle noise from the coherent imaging mechanism interfere strongly with the model to learn aircraft features. Therefore, by amalgamating the attention mechanism and node information derived from ST, ST-CAM is introduced. ST-CAM generates channel descriptors for each channel, thereby suppressing interference and amplifying the prominence of the aircraft, as shown in Fig. 6.

In ST-CAM, the node information of ST is represented as the set of scattering features in the critical regions of the aircraft. First, the set of node positions S is obtained from the SKPB as described above. Second, based on the 2-D coordinates of the node $s_i = (x_i, y_i)$, we obtain the feature vector $f_i \in R^C$ on $P_i{}'$, where C is the dimension of the $P_i{}'$ channel. The element z_j in f_i is the value of the corresponding position of the jth channel, where $j \in [0, C-1]$. The previous step is performed for all nodes and get the set of scattering features $F = \{f_1, f_2, \ldots, f_M\}$. Next, fusing all of the node information, $\widetilde{f}$ is acquired as

$$\widetilde{f} = F_{\text{sq}}(F) = \text{AvgP}(\text{Cat}(F)) \tag{3}$$

where $\text{Cat}(\cdot)$ denotes concatenates and $\text{AvgP}(\cdot)$ denotes average pooling. Specifically, the vectors in F are first concatenated and then compressed by using average pooling to obtain $\widetilde{f} \in R^C$. The element z_j in $\widetilde{f}$ is the descriptor of the jth channel. To capture the interchannel dependencies, c_{att} is obtained by

$$c_{\text{att}} = F_{\text{ex}}\left(\widetilde{f}\right) = \sigma\left(\text{FC}\left(\widetilde{f}\right)\right) \tag{4}$$

where $\text{FC}(\cdot)$ denotes the fully connected layer, σ refers to the ReLU [29] function, and $c_{\text{att}} \in R^C$. The final ST-CAM output P'' is obtained by scaling P' with c_{att}, as depicted in

$$P_i'' = \text{Scal}\left(P_i', c_{\text{att}}\right) \tag{5}$$

where $\text{Scal}(\cdot)$ refers to channel-wise multiplication between $P_i{}'$ and c_{att}.

We propose ST-CAM inspired by SENet [66]. However, the factors involved in calculating the channel descriptor are different. SENet uses all points in the spatial dimension, whereas ST-CAM uses only nodes in the ST. In other words, we utilize the scattering keypoint locations to sample the feature $P_i{}'$ prior to computation. Compared to nodes in the ST, the other locations in the spatial dimension contain more background information. And the importance of each channel depends on how much airplane information is included. Therefore, we employ this sampling method, enabling the model to direct its attention more prominently toward the aircraft, thereby mitigating extraneous background influences. And the increase in computational overhead is negligible.

D. Completeness and Consistency Loss

Various parts of the aircraft have different backscattering intensities, which leads to a significant discreteness of the SAR aircraft, resulting in predictions of missing parts. The existing loss functions are primarily designed for shape-continuous targets, rendering them unsuitable for discrete, multicenter SAR aircraft. The ST associates aircraft critical scattering regions that are significantly more important for precise aircraft regression compared to other areas within the bounding box. Therefore, CCLoss is proposed based on ST to enhance the positioning accuracy for discrete, multicenter SAR aircraft.

In CCLoss, the completeness-aware label is introduced to enable the model to measure the completeness of predictions.

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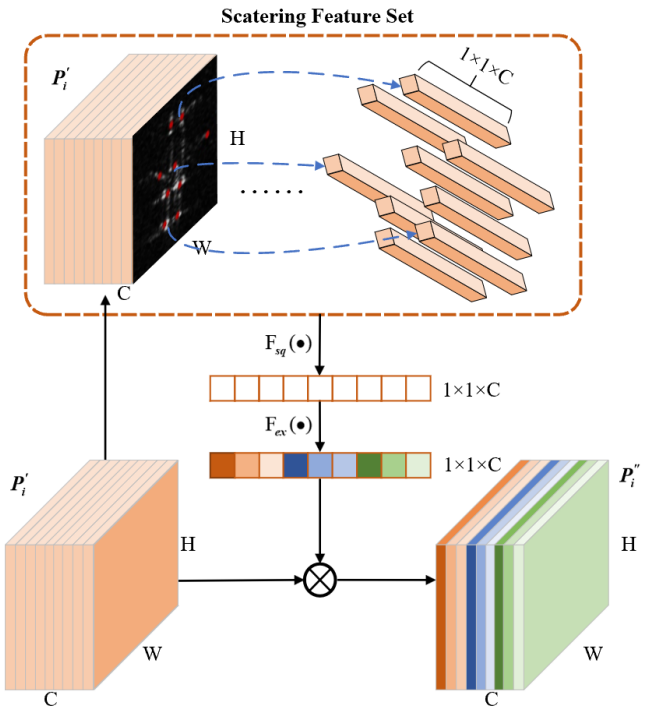
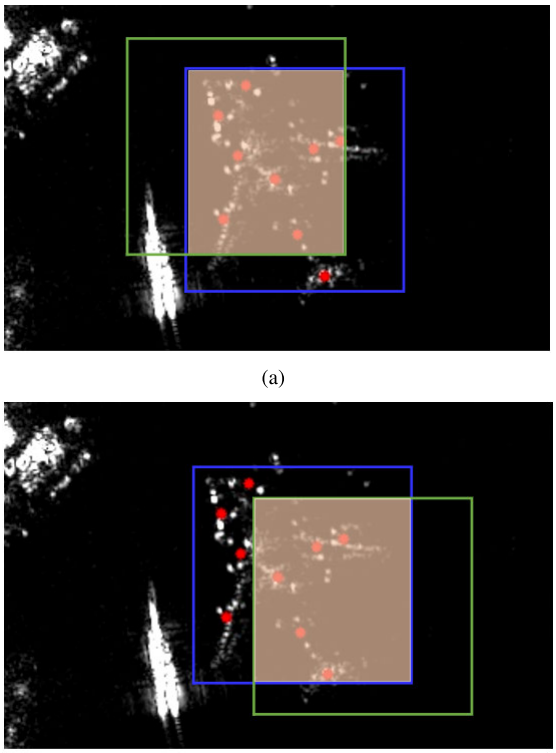


Fig. 6. ST-CAM. Based on the node positions, scattering feature set is derived from P_i', subsequently compressed and expanded to obtain the channel attention mask. This mask is then employed to emphasize the semantic information of aircraft.



(a)

(b)

Fig. 7. Comparison of prediction results. (a) Prediction Result A. (b) Prediction Result B. The blue and green boxes represent the ground truth bounding boxes and the predicted bounding boxes, respectively. The orange regions indicate the area of intersection between the predicted boxes and the ground truth boxes, while the red dots represent the key components of the aircraft. Although both results have the same IoU, they differ in their coverage of key aircraft components.

Moreover, most detectors commonly employ nonmaximum suppression (nms) to handle predictions, and the accuracy of the class score ordering significantly impacts the efficacy of nms. Therefore, the positive weighting function is incorporated into CCLoss to enhance the consistency between the classification and localization tasks.

1) Completeness-Aware Label: Most detectors employ IoU to quantify the completeness of prediction results, which is suitable for targets with continuous shape, but is not competent for discrete, multicenter SAR aircraft. We elaborate on the insufficiency of relying solely on IoU to assess the completeness of SAR aircraft, using Fig. 7 as an example. Compared to the former, the latter obviously lacks crucial components like the nose and wings, resulting in lower completeness. Nonetheless, both exhibit the same IoU, indicating that IoU fails to fully capture the disparity in completeness. Consequently for SAR aircraft, a suitable completeness criterion needs to satisfy the following conditions. First, it should be applied to discrete, multicentered targets, reflecting the scattering characteristics of SAR aircraft. Second, it should be universal to some extent. Specifically, the standard must be translation invariant and rotation invariant. Third, it should have simplicity and be easily embedded in the model. Considering the above conditions, we introduce $\overline{p^{\mathrm{sk}}}$ to contribute to the assessment of SAR aircraft integrity, which is the average confidence score of the first M points in the bounding box, as depicted in

$$\overline{p^{\mathrm{sk}}} = \frac{1}{M}\sum_{i=1}^{M} p_i^{\mathrm{sk}} \tag{6}$$

where p_i^{sk} is the confidence score used to measure whether the position is a strong scattering point or not. And the p_i^{sk} is obtained via SKPB. Obviously, the closer the number of scattering keypoints is to M in the bounding box, the more aircraft components it contains and the higher the completeness score it receives. In addition, $\overline{p^{\mathrm{sk}}}$ is independent of target position, and size. Furthermore, calculating $\overline{p^{\mathrm{sk}}}$ is based on the output of the SKPB and the regression branch, which costs only a small amount of computational resources and introduces no additional parameters.

Enhancing the model's capability to evaluate the completeness of results is crucial, as it enables the filtering out of incomplete results. However, if only using category labels to supervise model training is not sufficient for the above purpose, we illustrate with the example of the binary cross-entropy (CE) loss function, expressed as

$$\mathrm{CE}(p,q) = \begin{cases} -\log(p), & q = 1 \\ -\alpha\log(1-p), & q = 0 \end{cases} \tag{7}$$

where q indicates a category label and p denotes the predicted probability, and α is the moderator. Training the model using CE results in a classification score, which merely characterizes the probability that the prediction result is the target of interest. However, the localization accuracy of the prediction result cannot be directly inferred from the model output due to the unavailability of ground truth during model inference. In other words, the model has no sense of localization accuracy. Therefore, by combining the IoU and the $\overline{p^{\mathrm{sk}}}$, we propose

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the completeness-aware label q^{com} denoted as

$$q^{\text{com}} = q \times \frac{\overline{p^{\text{sk}}} + \eta \times \text{IoU}}{1+\eta} \tag{8}$$

where η is the moderator, which is used to modulate the contribution between $\overline{p^{\text{sk}}}$ and IoU. For positive samples, $(q = 1)$, q^{com} fuses category and completeness information; for negative samples $(q = 0)$, we are not concerned with their completeness thus remaining unchanged from before. In contrast to $q \in \{0, 1\}$, $q^{\text{com}} \in [0, 1]$ is a float number. To accommodate this change, the continuous CE loss function is used

$$\text{continuous CE}\left(p, q^{\text{com}}\right) = \begin{cases} -(q^{\text{com}} \log(p) + (1 - q^{\text{com}}) \log(1-p)), & q^{\text{com}} > 0 \\ -\alpha \log(1-p), & q^{\text{com}} = 0. \end{cases} \tag{9}$$

The q^{com}, which incorporates completeness scores, not only contains more information, but also takes into account the advantages of soft labels. For example, it reduces the risk of overfitting and can improve optimization efficiency [67], [68].

2) Positive Weighting Function: Consistency between model classification and regression entails that predictions exhibiting high classification scores also demonstrate high localization accuracy. And the localization accuracy is described by IoU. Instead, the model will get suboptimal detection results after nms. Specifically, predictions with high class scores but low localization accuracy might be retained, leading to false alarms. And predictions with low class scores but high localization accuracy might be filtered, resulting in missed targets. By analyzing the above two scenarios, we find that at least one of the values of confidence score and IoU is small in the inconsistent prediction results. It naturally follows that consistency is positively correlated with the product of confidence score and IoU, as depicted in

$$s^{\text{con}} = p \times \text{IoU} \tag{10}$$

where s^{con} denotes the consistency score, p denotes the predicted probability. Referring to the motivation of FL, samples with poor consistency are difficult samples and more attention needs to be given to them in training. Therefore, the positive weighting function is proposed for positive samples based on the consistency score s^{con}, as depicted in

$$w_p = \left(1 - s^{\text{con}}\right)^{\beta} \tag{11}$$

where β is the adjustment factor.

Combining the proposed completion-aware label and the positive weighting function, we propose CCLoss for classification branch, as depicted in the following equation:

$$\text{CCL}\left(p, q^{\text{com}}\right) = \begin{cases} -w_p(q^{\text{com}} \log(p) + (1 - q^{\text{com}}) \log(1-p)), & q^{\text{com}} > 0 \\ -\alpha p^{\mu} \log(1-p), & q^{\text{com}} = 0. \end{cases} \tag{12}$$

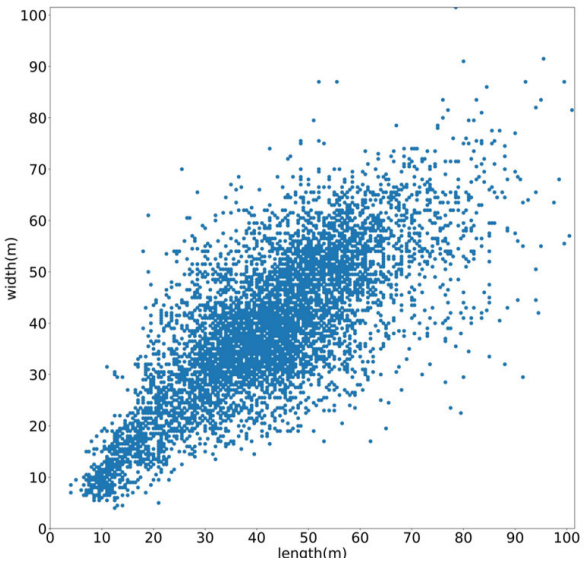


Fig. 8. Length and width distribution of the aircraft in GF3ADD.

E. Total Loss Function

The total loss function used to train STC-Net is divided into three parts

$$L = \lambda L_{\text{sp}} + L_{\text{cls}} + L_{\text{box}} \tag{13}$$

where L_{sp}, L_{cls}, and L_{box} represent scattering keypoint loss, classification loss, and localization loss, respectively, and λ is the loss modifier.

Scattering keypoint loss and classification loss are expressed as follows:

$$L_{\text{sp}} = \text{FL}\left(p^{\text{sk}}, q^{\text{sk}}\right) \tag{14}$$

$$L_{\text{cls}} = \text{CCL}\left(p, q^{\text{com}}\right). \tag{15}$$

For a comprehensive overview of L_{sp} and L_{cls}, see above. As for the localization task, we use IoU loss for supervision, denoted as follows:

$$L_{\text{box}} = 1 - \text{IoU}. \tag{16}$$

IV. Experiments

This section presents a series of experiments and analyzes the results to demonstrate the effectiveness of the proposed method. First, information about the GF3ADD dataset and details of the experiments are provided. Second, a brief description of the metrics used to evaluate the model performance is presented. Then, we construct ablation experiments to select appropriate hyperparameters and validate the role of proposed modules. Finally, experiments comparing the proposed model with other classical models are performed on dataset GF3ADD.

A. Dataset and Settings

1) GF3ADD: The 109 large scene images used to construct the dataset GF3ADD are from the Gaofen-3 satellite in spotlight imaging mode, which operates in the C-band with HH and VV polarization. These images have a nominal resolution

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of 1 m and cover large areas with complex backgrounds. 79 images are randomly selected for the training set, while 30 images are designated for the test set. Then, we crop each large image in a sliding window into 512 × 512 slices with the 200 pixel overlap. Dataset GF3ADD consists of 3519 slices obtained by cropping and contains 8129 aircraft targets. And the length and width distribution of the aircraft targets is shown in Fig. 8.

2) SAR-AIRcraft-1.0: The single-polarized, large-scale SAR images utilized to construct the SAR-AIRcraft-1.0 dataset [69] are acquired from the Gaofen-3 satellite, with a spatial resolution of 1 m. The SAR-AIRcraft-1.0 dataset comprises a total of 4368 images and 16 463 aircraft, categorized into four distinct sizes: 800 × 800, 1000 × 1000, 1200 × 1200, and 1500 × 1500 pixels. The dataset presents significant challenges due to the complexity of the scenarios and the considerable variability in aircraft scales. It should be noted that the challenge of minimal interclass differences coupled with substantial intraclass variability among aircraft targets in SAR images [51], [70] significantly complicates the differentiation between various types of aircraft. Without directly addressing the aforementioned challenges, the proposed method focuses primarily on accurately distinguishing aircraft from the background and achieving precise localization, which aligns with research paradigms that have been widely employed in related studies [5], [6], [57], [59], [60], [61], [63]. Consequently, in the subsequent experiments, we treat the aircraft targets in SAR-AIRcraft-1.0 dataset as a single class for detection.

Extensive experiments are conducted on GF3ADD and SAR-AIRcraft-1.0 to verify the effectiveness of the model. The parameters of the proposed module are initialized by a normal distribution. We set the batch size to 8 and train the model for 36 epochs with an initial learning rate of $1e^{-3}$, which drops to $1e^{-4}$ and $1e^{-5}$ at epoch 25 and 34, respectively. Stochastic gradient descent with a weight decay rate of $2e^{-4}$ and momentum of 0.9 is used as an optimizer.

B. Evaluation Metric

The performance of the model is represented using widely adopted metrics such as precision, recall, average precision (AP), and $F1$ score. As two intuitive evaluation metrics, precision and recall are, respectively, used to indicate the correctness of the prediction results and the completeness of the prediction results, which are given as (17) and (18), where TP refers to true positives, i.e., correct predictions, FP refers to false positives, i.e., predictions that are misclassified as objects, and FN refers to false negatives, i.e., objects that are not detected

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{18}$$

To balance precision and recall, they are consolidated and averaged to generate the $F1$ score value, as depicted in

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{19}$$

In general, the precision–recall (PR) curve reveals that the precision of the model decreases as recall rises. Therefore, to fully characterize the performance of the model under each recall–precision pair, AP is introduced and calculated as

$$\text{AP} = \int_0^1 P(R) dR. \tag{20}$$

The IoU describes the extent to what the predictions intersect with the ground truth, which is distributed from 0 to 1. When the IoU surpasses a predetermined threshold value, prediction results are defined as TP and vice versa. The AP50, obtained with the IoU threshold set to 0.5, is employed to characterize the model's capability to approximately localize the object. In addition, AP75 has been introduced to characterize the ability to accurately localize.

C. Ablation Studies

In this section, ablation experiments are conducted to verify the effect of the proposed modules in STC-Net. We denote the preliminary networks mentioned in Section III-A as baseline. Based on the baseline, STFFM and CCLoss are added separately to investigate the role for model performance improvement. For a fair comparison, the settings remain consistent across all experiments. The results of ablation experiments is shown in Table I. It is obvious that each proposed module contributes to the enhancement of the model's performance to some extent. In the end, the STC-Net outperform the baseline over 3.4% in AP50 and 4.6% in AP75. We describe in detail the performance improvements brought by each proposed module.

1) Effect of STFFM: Based on the ST, we construct the STFFM to obtain more discriminating features. STFFM consists of ST-SFM and ST-CAM, where the former is utilized to aggregate discrete aircraft features and the latter is employed to suppress background interference. We integrate the two sub-modules separately to the baseline to see their impact on the model, and the results are shown in Table II. By comparing the first two rows of Table II, it is observed that ST-SFM improves AP50 and AP75 by 1.6% and 1.5%, respectively. And by comparing the first row with the third row in Table II, it is seen that the ST-CAM improves AP50 and AP75 by 1.2% and 1.8%, respectively. The results indicate that both sub-modules enhance the model's detection capability in distinct manners. ST-SFM enhances the association between the various components of the aircraft based on the neighboring information of ST. And, ST-CAM combines the attention mechanism with node information of ST to make the network more focused on aircraft targets.

In addition, we investigate the effect of various connection mode among these two submodules, including one parallel and two serial mode. ST-CAM precedes ST-SFM in Series I and the reverse Series II, as shown in Table III. The optimality of Series II among the three connections is evident. Eventually, STFFM cascades with ST-CAM to form ST-SFM, which improves AP50 and AP75 by 1.6% and 3.3%, respectively.

Furthermore, the comparison of the prediction results is shown in Fig. 9. We can find that there are some strong scattering targets in the SAR images that have similar intensity

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TABLE I
PERFORMANCE OF EACH COMPONENT IN STC-NET ON THE GF3ADD

STFFM	CCLoss	AP50	AP75	Recall	Precision	F1	Params(M)	Time(ms)
×	×	0.8899	0.4370	0.8737	0.8904	0.8819	**31.84**	**21.6**
✓	×	0.9061	0.4702	0.8921	0.8991	0.8956	35.41	27.2
×	✓	0.9163	0.4660	0.8866	0.896	0.8913	34.76	24.8
✓	✓	**0.9241**	**0.4833**	**0.8925**	**0.9037**	**0.8981**	35.41	29.6

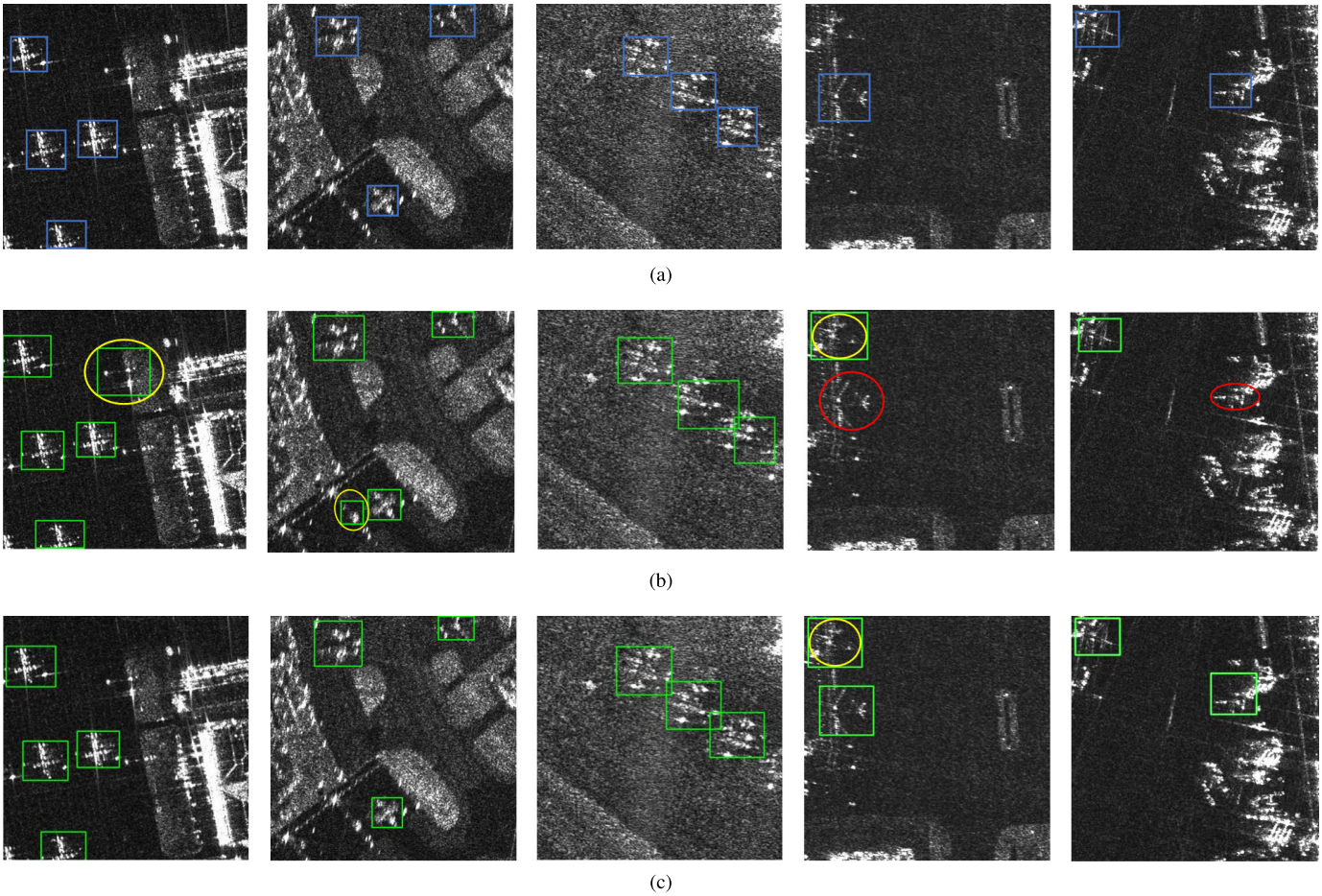


Fig. 9. Detection results of the baseline without and with STFFM. (a) Baseline. (b) Baseline with STFFM. (c) Ground truth of the images. The blue and green boxes represent ground truth and detection results, respectively, while the yellow and red circles represent false alarms and aircraft that are missed, respectively.

TABLE II
PERFORMANCE OF EACH COMPONENT IN STFFM

ST-SFM	ST-CAM	AP50	AP75	Recall	Precision	F1
×	×	0.8899	0.4370	0.8737	0.8904	0.8819
✓	×	0.9055	0.4516	0.8879	0.8936	0.8907
×	✓	0.9016	0.4548	0.8807	0.8966	0.8886
✓	✓	**0.9061**	**0.4702**	**0.8921**	**0.8991**	**0.8956**

TABLE III
INFLUENCE OF SUBMODULE CONNECTIVITY IN THE STFFM

Connection Mode	AP50	AP75	Recall	Precision	F1
Parallelism	0.9029	0.4629	0.8848	0.8913	0.8880
Series I	0.9058	0.4662	0.8802	0.8991	0.8895
Series II	**0.9061**	**0.4702**	**0.8921**	**0.8991**	**0.8956**

and texture characteristics with the airplane. The baseline tends to detect them as airplanes thus causing false alarms. Moreover, the baseline misses the aircraft when the aircraft scattering intensity is weak and there is a man-made structure with strong scattering in the background such as a corridor bridge. In addition, neighboring aircraft can interfere with each other, resulting in inaccurate positioning of the baseline. STFFM enhances the integrity and saliency of the aircraft by aggregating discrete structures within the aircraft and suppressing strong interference. Thus, the baseline with STFFM can better cope with the above problems.

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TABLE IV
INFLUENCE OF COMPLETENESS-AWARE LABEL MODULATOR η

η	AP50	AP75	Recall	Precision	F1
0	0.8849	0.4390	0.8527	0.8420	0.8473
1	0.8881	0.4466	0.8625	0.8816	0.8719
2	0.8890	0.4470	0.8720	0.8849	0.8784
3	**0.9086**	**0.4557**	**0.8764**	**0.8921**	**0.8842**
4	0.8964	0.4550	0.8719	0.8825	0.8772
5	0.8948	0.4422	0.8605	0.8601	0.8603
∞	0.8974	0.4320	0.8491	0.8625	0.8557

TABLE V
INFLUENCE OF THE POSITIVE WEIGHTING MODERATOR β

β	AP50	AP75	Recall	Precision	F1
1	**0.9133**	**0.4517**	**0.8856**	**0.8943**	**0.8899**
2	0.9076	0.4450	0.8838	0.8871	0.8854
3	0.9076	0.4511	0.8834	0.8942	0.8888
4	0.9032	0.4447	0.8779	0.8913	0.8845

The results show that STFFM improves positioning accuracy and reduces false alarms by completing the aircraft geometry and suppressing background interference.

2) Effect of CCLoss: CCLoss contains two main parts: completeness-aware label and positive weighting function. The former allows the model to perceive the completeness of the samples during training, while the latter allows the model to focus on difficult samples, i.e., samples whose classification does not agree with the regression. We analyze the role they play in the model in order.

The completeness-aware label contains hyperparameter η to balance the IoU with the $\overline{p^{\mathrm{sk}}}$. As η grows, the contribution of IoU in it increases. In order to select the appropriate value, we designed relevant experiments for η, as shown in Table IV. A decrease in model performance compared to the baseline can be observed when η is set to 0. This means that just using $\overline{p^{\mathrm{sk}}}$ to describe the completeness of the aircraft would instead be somewhat misleading to the model. Similarly, when η is set to ∞, the change of model performance is very marginal. This means that using only IoU to characterize the integrity of an aircraft is also insufficient. And the model achieves the best performance when η is set to 3. This suggests that a completeness description that includes scattering features is more applicable to SAR aircraft detection.

Additionally, the effect of the positive weighting function, which incorporates a hyperparameter β, is analyzed. Correlation experiments are conducted to assess its impact, as depicted in Table V. When β is set to 0, the positive weighting function does not work, thereby resulting in the degeneration of the loss function to the CE shown in Section III-D. Moreover, the model achieves optimal performance when β is set to 1. Upon hyperparameter selection, CCLoss is incorporated into the baseline methodology. Table I illustrates the outcomes, revealing a notable enhancement of 2.6% in AP50 and 2.9% in AP75 relative to the baseline. Furthermore, the comparison

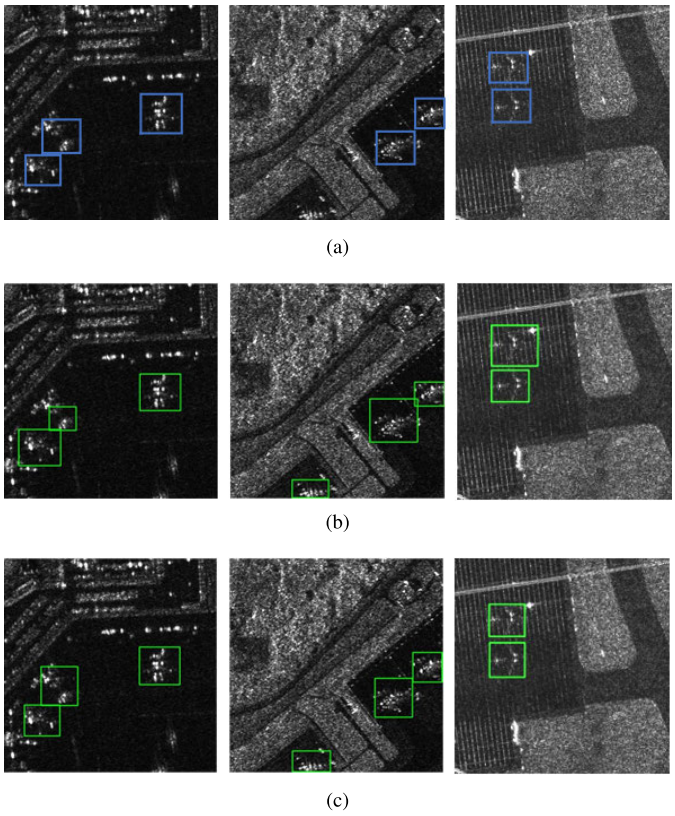


Fig. 10. Detection results of the baseline without and with CCLoss. The green boxes represent detection results. (a) Baseline. (b) Baseline with CCLoss. (c) Ground truth of the images.

of prediction results is depicted in Fig. 10. We can see that when there is strong scattering interference around the aircraft target, the baseline predicts a larger detection box that includes both the aircraft and the strong scattering interference. In addition, the prediction results of the baseline will miss critical components such as the nose when the aircraft is highly discrete. Conversely, the model trained using CCLoss performs better in the above scenarios. Without introducing additional parameters, CCLoss allows the model to better perceive the completeness of the prediction results, thus improving the localization accuracy.

3) Scattering Keypoint Extraction: Since the scattering keypoints are the base element of the ST, we analyze the methods for extracting the scattering keypoints in the end. We set up three extraction ways: the Candy edge detection, the Harris corner point detection, and the use of both. The results are shown in Table VI and it can be concluded that using both edge and corner point detection is optimal. We can see that the Candy algorithm obtains the contours of the aircraft and the Harris algorithm focuses on the prominent structures in the aircraft. There is some degree of intersection between the two types of information, but neither provides a comprehensive description of the structure of the aircraft. Therefore, using both results can complement each other, obtaining more complete information about the aircraft geometry in order to get the optimal results.

4) Loss Modifier λ: The λ is the hyperparameter used to control the influence of the scattering keypoint loss on the

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TABLE VI
IMPACT OF METHODS FOR EXTRACTING SCATTERING KEYPOINTS

Candy	Harris	AP50	AP75	Recall	Precision	F1
×	×	0.8899	0.4370	0.8737	0.8904	0.8819
✓	×	0.9138	0.4781	0.8889	0.9004	0.8946
×	✓	0.9139	0.4768	0.8857	0.8997	0.8926
✓	✓	**0.9241**	**0.4833**	**0.8925**	**0.9037**	**0.8981**

TABLE VII
INFLUENCE OF THE LOSS MODIFIER λ

λ	AP50	AP75	Recall	Precision	F1
0.5	0.9176	0.4754	**0.8975**	0.9033	**0.9004**
1	**0.9241**	**0.4833**	0.8925	0.9037	0.8981
2	0.9184	0.4768	0.8916	**0.9069**	0.8992

TABLE VIII
PERFORMANCE COMPARISON BETWEEN DC AND STFFM

DC	ST-FFM	AP50	AP75	Recall	Precision	F1
×	×	0.8899	0.4370	0.8737	0.8904	0.8819
✓	×	0.9033	0.4358	0.8752	0.8901	0.8826
×	✓	0.9055	0.4516	0.8879	0.8936	0.8907

TABLE IX
PERFORMANCE COMPARISON BETWEEN CAM AND ST-CAM

CAM	ST-CAM	AP50	AP75	Recall	Precision	F1
×	×	0.8899	0.4370	0.8737	0.8904	0.8819
✓	×	0.8968	0.4313	0.8541	0.9042	0.8784
×	✓	0.9016	0.4548	0.8807	0.8966	0.8886

overall model loss. Experiments are conducted to determine its optimal value, with the results presented in Table VII. The results indicate that the model performance remains relatively stable with λ values of 0.5, 1, and 2, with the best performance achieved at $\lambda = 1$.

D. Discussion

1) ST Guidance: ST-SFM and ST-CAM, designed based on the neighboring and node information of the ST, respectively, both effectively enhance the model's performance. To thoroughly validate the role of ST guidance, relevant experiments are designed.

In ST-SFM, the neighboring information of the ST is used to refine the initial offsets, and the refined offsets are then used to guide the DC. To validate the role of neighboring information, we omit the offset refinement process and instead use a standard DC to replace ST-SFM. The experimental results in Table VIII indicate that the introduction of DC results in a 1.3% improvement in AP50, while other metrics remain nearly unchanged. In contrast, ST-SFM achieves superior performance, particularly in AP75. These results imply that without refining the initial offset using neighboring information, DC faces challenges in achieving optimal results through implicit learning.

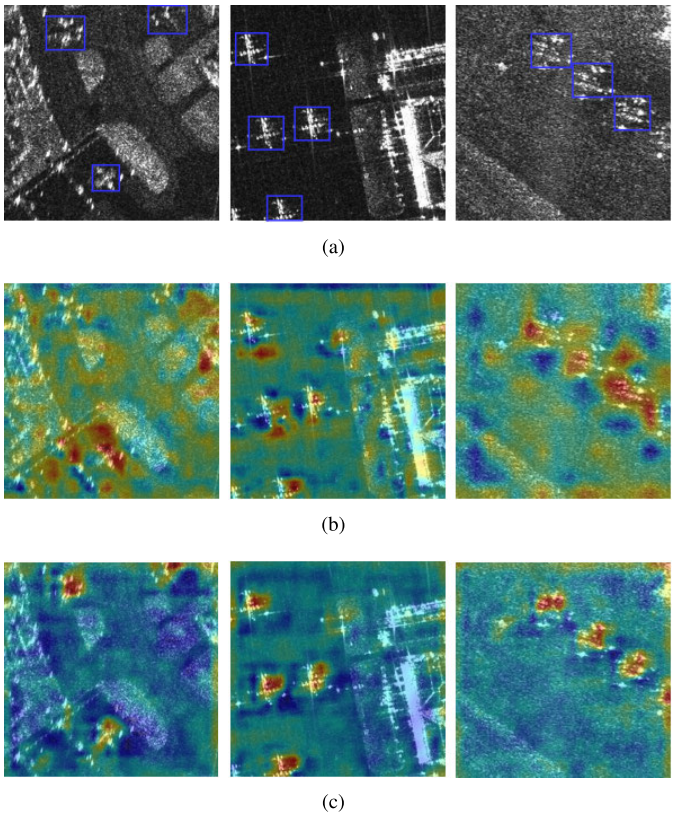


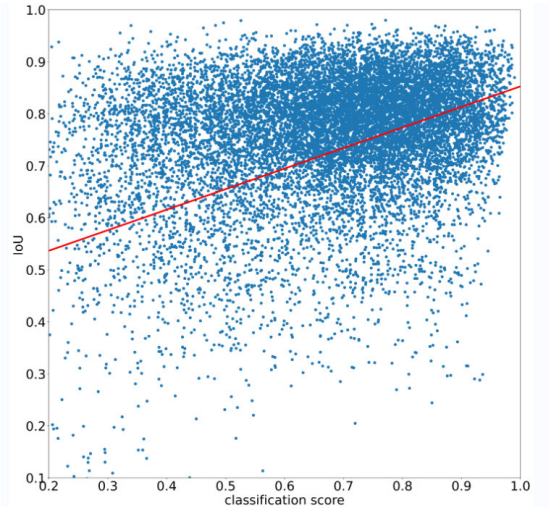
Fig. 11. Visualization of feature maps extracted by different methods. (a) Ground truth of the images. (b) Method without STFFM. (c) Method with STFFM.

In ST-CAM, the node information of ST is compressed and interacted to generate a channel description, which is used for feature map scaling. To verify the role of node information, the node feature set is replaced with the entire feature map to generate the channel descriptor, which causes the module to degrade into a standard CAM. The experimental results are presented in Table IX. It can be observed that after introducing CAM, AP50 and precision increased by 0.7% and 1.4%, respectively, while AP75 and recall decreased by 0.6% and 1.8%, respectively. In contrast, after introducing ST-CAM, all metrics of the model improved comprehensively. This is because the entire feature map contains substantial background information, while the node feature set has a higher proportion of aircraft semantic information. In other words, the channel descriptor obtained by compressing and interacting with the node feature set can better reflect the distribution of aircraft semantic information within the feature map.

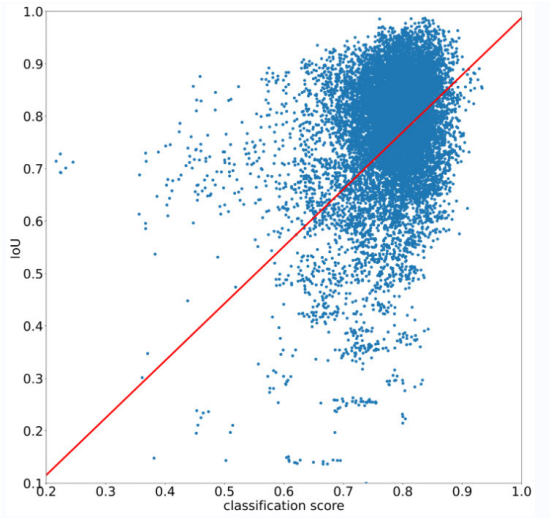
In summary, ST offers more interpretable information guidance and avoids the implicit learning paradigm of standard convolutions, enabling both STFFM and ST-CAM to achieve superior performance.

2) STFFM: Along with some visualized images, as shown in Fig. 11, we qualitatively analyze the role of STFFM plays in the model. The extracted feature map F is compressed by summing over the channel dimension and then normalized to the range from 0 to 1 to obtain the activation heatmap, which is subsequently overlaid with the original image to generate the visualized image. As shown in Fig. 11(b), when there are some strongly scattering artificial buildings in the background,

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(a)



(b)

Fig. 12. Classification score-IoU distribution plots and linear fitting results for different methods. (a) Method without CCloss. (b) Method with CCloss.

they tend to be highly corresponding in the feature map as well. Moreover, neighboring aircraft interfere with each other, blurring the boundaries of each other's responses on the feature map. In addition, the response of the aircraft itself shows a discrete distribution. The feature map with these problems present significant challenges for downstream classification and localization tasks. As shown in Fig. 11(c), the background interference is significantly suppressed and the boundaries of neighboring aircraft responses are more clearly defined with the help of STFFM. At the same time, the response of the aircraft is more focused. The comparison demonstrates that STFFM makes aircraft features more discriminative. We argue that this enhancement is due to two factors. One is that more flexible sampling aggregates the internal structure of the aircraft, and the other is that the channel descriptors based on the ST suppress strong scattering interference.

3) CCLoss: To qualitatively analyze the role of CCLoss, we constructs a planar space with the classification score as the horizontal axis and IoU as the vertical axis. We compute

TABLE X
COMPARISON OF DIFFERENT CNN-BASED METHODS ON GF3ADD

Method	AP50	AP75	Recall	Precision	F1
SSD [39]	0.8777	0.3885	0.8139	0.8549	0.8339
Faster RCNN [33]	0.9038	0.4614	0.8244	0.8413	0.8328
Cascade RCNN [36]	0.9065	0.4623	0.8313	0.8409	0.8361
FCOS [47]	0.9021	0.4336	0.8518	0.8978	0.8742
FoveaBox [48]	0.9035	0.4538	0.8573	0.8363	0.8467
FSAF [71]	0.9006	0.4390	0.8802	0.8773	0.8787
GFL [72]	0.9151	0.4641	0.8737	0.8778	0.8757
PAFPN [73]	0.8866	0.4623	0.8724	**0.9137**	0.8926
RepPoint [74]	0.8874	0.4501	0.8472	0.8943	0.8701
SABL [75]	0.9107	0.4653	0.8541	0.9143	0.8832
Sparse RCNN [76]	0.8768	0.4709	0.8587	0.8955	0.8767
VFNet [64]	0.9154	0.4666	0.8596	0.8973	0.8780
STC-Net	**0.9241**	**0.4833**	**0.8925**	0.9037	**0.8981**

TABLE XI
COMPARISON OF DIFFERENT CNN-BASED METHODS ON SAR-AIRCRAFT-1.0

Method	AP50	AP75	Recall	Precision	F1
SSD [39]	0.8496	0.4563	0.7854	0.8986	0.8382
Faster RCNN [33]	0.7160	0.5360	0.7810	0.7760	0.7785
Cascade RCNN [36]	0.7780	0.5910	0.7950	0.8900	0.8398
FCOS [47]	0.8569	0.4610	0.8162	0.8631	0.8390
FoveaBox [48]	0.8312	0.4525	0.8018	0.8801	0.8391
FSAF [71]	0.8344	0.4531	0.8142	0.8696	0.8410
GFL [72]	0.8472	0.5493	0.8176	0.8648	0.8405
PAFPN [73]	0.8065	0.5136	0.8335	0.8432	0.8383
RepPoint [74]	0.8030	0.5290	0.8870	0.6270	0.7347
SABL [75]	0.8554	0.5163	0.8305	0.8915	0.8599
Sparse RCNN [76]	0.7857	0.4941	0.6989	0.8166	0.7532
VFNet [64]	0.8279	0.5673	0.7345	0.8699	0.7965
SKG-Net [77]	0.7980	0.5100	**0.8880**	0.5760	0.6988
SA-Net [69]	0.8040	0.6140	0.8220	0.8750	0.8477
STC-Net	**0.8898**	**0.6197**	0.8406	**0.9048**	**0.8715**

the classification score and IoU of the predictions and plot them into classification score-IoU space, as shown in Fig. 12. It can be observed that the results predicted by the model with CCLoss are more centrally distributed in the upper right corner, compared to model without CCLoss. This suggests that more predictions have both higher classification score and localization accuracy. In addition, we obtain the linear fit function of points using the least squares method as depicted in the following equations:

$$\mathrm{IoU} = 0.4 \cdot \mathrm{cls_score} + 0.46 \tag{21}$$

$$\mathrm{IoU} = 1.09 \cdot \mathrm{cls_score} - 0.1. \tag{22}$$

Obviously, classification scores of the results obtained from the method with CCLoss better reflects the completeness of the prediction results. This shows that CCLoss enhances the

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consistency of classification and localization branches, which provides a solid foundation for subsequent nms operations.

E. Comparison With CNN-Based Methods

To substantiate the superiority of the proposed model, we conduct a comparative analysis against other CNN-based detectors, employing a series of experiments on the GF3ADD dataset. To ensure a comprehensive comparison, a total of 13 methods are examined, including SSD, Faster R-CNN, and FoveaBox. These methods cover not only anchor-based and anchor-free detectors, but also single-stage and two-stage detectors. The comparison results are presented in Table X. Notably, the proposed method demonstrates optimality across various evaluation criteria including recall, $F1$ score, AP50, and AP75, which means that STC-Net can effectively highlight aircraft features and achieve more accurate localization. Meanwhile, compared to some lightweight detectors, the performance of STC-Net exhibits a substantial enhancement at the cost of a small additional computational overhead. This means that our model achieves a compromise between accuracy and speed.

Additionally, we conduct experiments comparing STC-Net with the aforementioned methods using the publicly available SAR-Aircraft-1.0 dataset to evaluate the generalization and robustness of the proposed method. Algorithms based on the scattering keypoints in SAR images, such as SKG-Net [77] and SA-Net [69], are also included in the comparative analysis. The experimental results, as detailed in Table XI, demonstrate that STC-Net achieves optimal performance. Specifically, when compared to SKG-Net, STC-Net achieves a 39.8% improvement in precision, albeit with a 4.8% decrease in recall. In comparison to SA-Net, STC-Net demonstrates a 8.5% enhancement in coarse positioning capability (AP50) and a modest increase in fine positioning capability (AP75). This indicates that the proposed method leverages the aircraft's scattering features and their topological relationships more effectively, leading to enhanced performance.

V. Conclusion

In this article, an STC-Net is proposed for aircraft detection in SAR images. Given the discreteness of the aircraft, we represent it using the ST, which contains a wealth of structural information. Furthermore, the SSFFM based on the ST enhances aircraft semantic information and suppresses background interference by dynamically aggregating and highlighting aircraft features. Furthermore, we introduce CCLoss to facilitate the model training process, thereby enhancing the model's capacity to perceive the completeness of the prediction results. Ablation experiments demonstrate that the proposed modules all play an important role, outperforming the baseline over 3.4% in AP50 and 4.6% in AP75. Comparison with other detectors on the GF3ADD and SAR-Aircraft-1.0 dataset demonstrates the superiority of our method.

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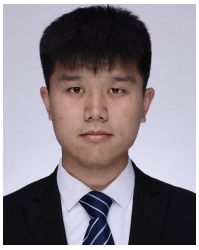
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